Experiment 9 FAQs

Frequently Asked Questions:

1. What is a tree?

Ans: Tree is a nonlinear data structure used for hierarchical data management.

2. What is a binary tree?

Ans: A binary tree is an empty tree or it is a finite set of nodes with root and two disjoint binary trees called left subtree and right subtree.

3. What are the types of Binary trees?

Ans: Complete binary tree,Full binary tree,binary search tree, Skewed binary tree ,threaded binary tree etc.

4. What are tree traversal techniques?

Ans: 1)Recursive 2)Non recursive

5. What are recursive tree traversal techniques?

Ans: 1)Inorder 2)Preorder 3)Postorder

6. What are AVL trees?

Ans: AVL trees are height balanced trees.

7. What are the basic operations that can be performed with tree?

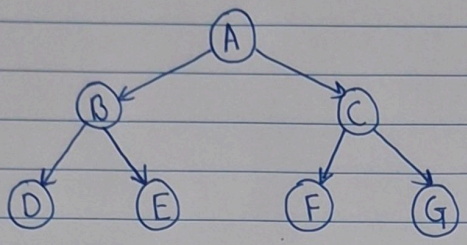
Ans: 1)Create 2)Search3)Traversal etc.

8. Explain the term degree and height of the tree.

Ans: Degree of tree : The number of subtrees of a node is called as its degree. Height or depth of tree: It is the maximum level of any node in the tree.

9. Draw representation of tree using Linked List and array.

Ans. 1) Array Representation



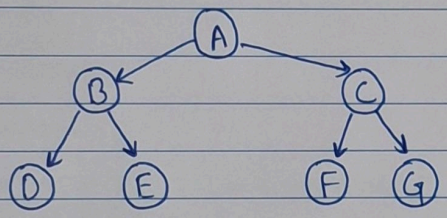
i) Stores tree nodes in a sequential array.
ii) Easy access using index.
iii) Suitable for complete (binary) trees.

Array,

Index :

0	1	2	3	4	5	6	
A	B	C	D	E	F	G	

2) Linked list Representation



i) Uses dynamic memory (nodes connected by pointers).
ii) Flexible size (can grow/shrink).
iii) Suitable for general binary trees.

For node A,

